NCP affine normal form: Lean proof and archive replay

2026-10-21; THM-COMMUTATOR-KERNEL1 Lemma 1; TASK-20260822-a0e34f

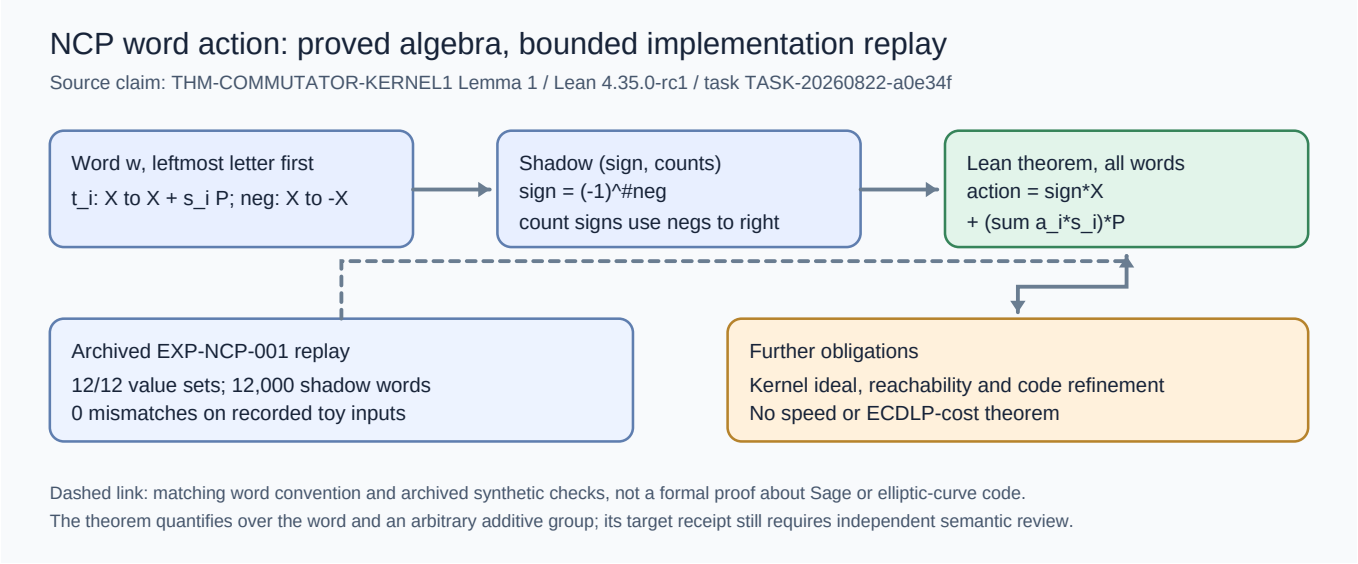
Scope. The new Lean file formalizes the word-action normal form in the archived translation/negation quiver. It does not prove the kernel ideal, reachability formula, elliptic-curve code refinement, or a speed result. No new scientific trial or timing measurement was made.

Statement and proof

Words apply letters from left to right. The shadow sign is $(-1)^{\#\nu}$; each translation's signed count uses the number of negations to its right. For scalars s_i in $\mathbb{Z}/n$, the checked theorem states, for every word w and starting scalar x,

$$\mathrm{action}_w(x) = \varepsilon(w)x + \sum_i a_i(w)s_i.$$

The specialization at $x = 1$ is the orbit-value law. A second theorem proves the stronger group form for translations s_iP in any additive commutative group: $\mathrm{action}_w(X) = \varepsilon(w)X + (\sum_i a_i(w)s_i)P$. Equal shadows induce identical actions. Induction on the word proves these identities: a translation adds a signed term and a negation flips the sign of the earlier translations.



Checks and evidence boundary

Check	Observed result
Pinned Lean workspace	Lake build passed 2,012 jobs on Lean 4.35.0-rc1; project axiom audit passed 484 theorem declarations with only standard axioms.
Frozen NCP target	Machine verified; source/task/toolchain/manifest hashes and source commit bound in the receipt. Independent semantic review remains pending.
Complete target corpus	63 machine verified, zero failed, zero infrastructure failures, three theorem files absent. Every present target still needs semantic review.
Archived NCP replay	Twelve of twelve frozen value sets matched; 12,000 shadow words had zero mismatches. The strict-inclusion row remains 350/439 versus 437/439. No fresh performance measurement.

The Lean statement concerns the algebraic action of words. The scratch replay rechecks the frozen affine rule and stored counts; the old sampled point arithmetic positive control was not rerun. Neither establishes correctness of all elliptic-curve implementation paths or a search-cost theorem. The two other frozen NCP targets (reachability and commutator-ideal refutation) remain without Lean files. No canonical numerical graph or dashboard changed.